

Homework 2

ASEN 5335 Aerospace Environment

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Problem 1: Spaceballs

(i.)

We are able to use the provided MSISatmosphere1000.m profile to calculate the an approximate total mass of the atmosphere. The MSISatmosphere1000 function takes inputs of altitudes and outputs a structure, one of which is mass density which corresponds to that altitude. Using MATLAB we are able to have a vector of altitudes that range to 1000 km, the maximum model input of the MSISatmosphere1000 function. We can then obtain a vector of mass densities ρ . We use the following,

$$m = \rho V$$

$$m = \int \rho dV$$

And for a sphere,

$$m = \int \rho * 4\pi r^2 dr$$

$$\therefore m = 4\pi \int \rho r^2 dr$$

With ρ being a vector and r also being a vector of radius of this atmospheric shell (altitude including radius of the Earth $R_E = 6371$ km), we can integrate within MATLAB to obtain a

$$\boxed{m = 5.297 * 10^{18} \text{ kg}}.$$

(ii.)

Assuming a volume of $V = 10 * 10^3 * 10 * 10^3 * 100 * 10^3 = 1 * 10^{13} \text{ m}^3$, a constant temperature $T = 300 \text{ K}$, molar mass $M = 0.029 \text{ kg/mol}$, and universal gas constant $R = 8.314 \text{ J/(molK)}$. We can use the ideal gas law,

$$PV = \frac{m}{M}$$

Rewrite to solve for pressure P ,

$$P = \frac{m}{MV} RT$$

$$\boxed{P = 4.556 * 10^{10} \text{ Pa}}$$

Problem 2: Atmospheric Scale Height

To calculate the scale height H of air mixture assuming N_2/O_2 mixture at sea level, we use the equation.

$$H = \frac{kT}{mg}$$

Where we assume a constant $T = 300$ K from the problem, and $k = 1.38 * 10^{-23}$ J/K, the Boltzmann constant, $g = 9.81$ m/s², the gravitational constant. We then seek the molecular mass of N_2/O_2 mixture. This can be obtained from the table provided in the HW. We find the atomic mass of the air mixture provided with the % by volume

$$amu_{air} = 28.0134 (.781) + 31.9988 (.209) = 28.566 \text{ g/mol}$$

$$m_{air} = amu_{air} * \frac{1 \text{ kg}}{1000 \text{ g} * 6.02 * 10^{23} \text{ g/mol}}$$

$$m_{air} = 4.745 * 10^{-26} \text{ kg}$$

This process of finding molecular weight and converting it can be repeated for atomic Oxygen, helium, and hydrogen.

$$amu_{oxygen} = (31.9988 \text{ g/mol})/2 = 15.9994 \text{ g/mol}$$

$$amu_{helium} = 4.0026 \text{ g/mol}$$

$$amu_{hydrogen} = 2.01594 \text{ g/mol}$$

$$m_{oxygen} = 2.6577 * 10^{-26} \text{ kg}$$

$$m_{helium} = 6.6488 * 10^{-27} \text{ kg}$$

$$m_{hydrogen} = 3.3487 * 10^{-27} \text{ kg}$$

The scale height equation can be used with each respective molecular mass of the species in question.

$$H_{air} = 8.8938 * 10^3 \text{ m}$$

$$H_{oxygen} = 1.588 * 10^4 \text{ m}$$

$$H_{helium} = 6.347 * 10^4 \text{ m}$$

$$H_{hydrogen} = 1.2602 * 10^5 \text{ m}$$

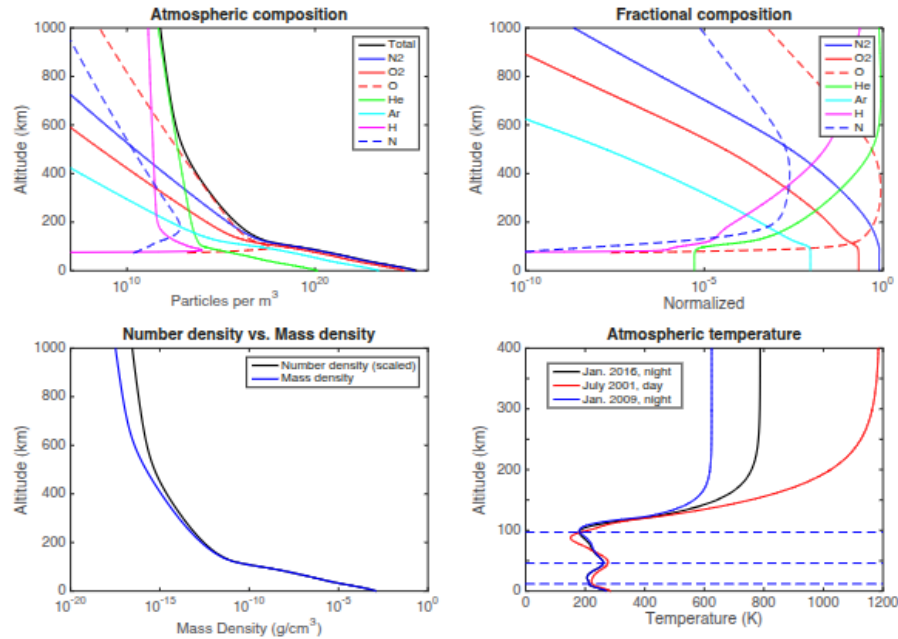


Figure 1: Atmospheric Profile

By comparing these scale heights to those in the Figure 1 provided from the HW, we observe that
 TODO: We seek to also calculate scale height from the MSIS profiles. From the MSIS profiles there will be different temperatures instead of the fixed T used earlier. Using altitudes of 150,180,450,500,600,1000km.

Problem 3: Atmosphere Variation

(i.)

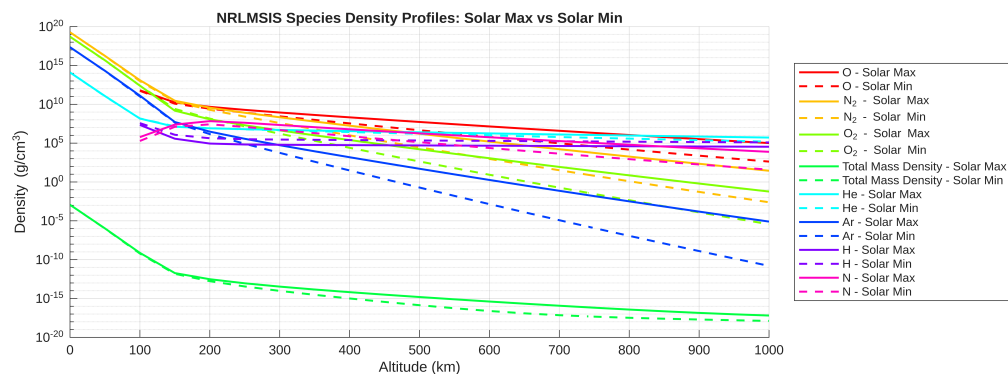


Figure 2: NRLMSIS Species Density Profiles Solar Max vs Solar Min

(ii.)

At 600 km, there is variation between the solar maximum and solar minimum with the density profiles of each species deviating. At solar maximums, the densities of each species tends to be larger by multiple orders of magnitude than they would be at solar minimums. This can even be seen overall with the Total Mass Density variation between solar maximums and minimums. The implication of this for spacecraft operations is that there will be different atmospheric conditions for spacecraft depending on if it is at a solar maximum or minimum. For instance, drag will be dependent on the density of the atmosphere, during solar maximums where density can increase by multiple orders of magnitude, drag will also be considerably affected. This could especially impact LEO orbits with their orbits being considerably influenced by drag. Their satellite operations could therefore vary depending on if it is at a solar maximum or minimum.

(iii.)

The dominant atmospheric constituent would be O, atomic oxygen, for both solar maximum and solar minimum. (TODO: Double check this...)

Problem 4: Oxygen Impact

(i.)

To find the characteristic velocity of an oxygen atom O at $T = 1000\text{ K}$ in the thermosphere, we are able to use the Maxwell Boltzmann Characteristic RMS Velocity equation.

$$v_{rms} = \sqrt{\frac{3k_B T}{m}}$$

$k_B = 1.38 * 10^{-23} \text{ J/K}$ as the Boltzmann constant, and we are able to find $m = 2.657 * 10^{-26} \text{ kg}$ for an oxygen atom. We can plug this into the equation to simply find,

$$v = 1248.26 \text{ m/s}$$

(ii.)

We then calculate the velocity that a spacecraft in LEO at 600 km would have,

$$\begin{aligned} v_{orbit} &= \sqrt{\frac{GM_{Earth}}{r}} \\ G &= 6.67 * 10^{-11} \text{ Nm}^2/\text{kg}^2 \\ M_{Earth} &= 5.972 * 10^{24} \text{ kg} \\ r &= 600 * 10^3 \text{ m} + 6371 * 10^3 \text{ m} = 6971 * 10^3 \text{ m} \\ \therefore v_{orbit} &= 7559 \text{ m/s} \end{aligned}$$

The energy imparted onto the spacecraft by an oxygen atom would be using the mass of the oxygen atom and the velocity of the spacecraft,

$$KE = \frac{1}{2} m_{oxygen} v_{orbit}^2$$

$$KE = 7.591 * 10^{-19} J$$

$$\text{Where } 1 J = 6.242 * 10^{18} \text{ eV}$$

$$\boxed{\therefore KE = 4.738 \text{ eV}}$$

(iii.)

TODO: TODO

Problem 5: Orbit Decay

(i.)

We use the following relations:

$$F_{drag} = \frac{1}{2} \rho V^2 C_D A, \quad V = \sqrt{\frac{GM_E}{a}}, \quad P^2 GM_E = 4\pi^2 a^3$$

Step 1: Drag equation

$$m \frac{dV}{dt} = \frac{1}{2} \rho V^2 C_D A$$

$$\frac{dV}{dt} = \frac{\rho V^2 C_D A}{2m}$$

Step 2: Relate da/dt to dV/dt with the velocity equation

$$V^2 a = GM_E$$

$$\left(\frac{d}{dt} \right) (V^2 a) = \left(\frac{d}{dt} \right) (GM_E)$$

$$2Va \frac{dV}{dt} + V^2 \frac{da}{dt} = 0$$

$$\frac{da}{dt} = -\frac{2a}{V} \frac{dV}{dt}$$

Substituting Step 1,

$$\frac{da}{dt} = -\frac{2a}{V} \left(\frac{\rho V^2 C_D A}{2m} \right)$$

$$\frac{da}{dt} = -\frac{a \rho V C_D A}{m}$$

Step 3: Relate da/dt to dP/dt via Kepler's third law

$$P^2 GM_E = 4\pi^2 a^3$$

$$\left(\frac{d}{dt}\right) (P^2 GM_E) = \left(\frac{d}{dt}\right) (4\pi^2 a^3)$$

$$2P GM_E \frac{dP}{dt} = 12\pi^2 a^2 \frac{da}{dt}$$

$$\frac{da}{dt} = \frac{P GM_E}{6\pi^2 a^2} \frac{dP}{dt}$$

Step 4: Combine

Setting the two expressions for da/dt equal,

$$-\frac{a\rho V C_D A}{m} = \frac{P GM_E}{6\pi^2 a^2} \frac{dP}{dt}$$

$$\frac{dP}{dt} = -\frac{6\pi^2 a^3 \rho V C_D A}{m P GM_E}$$

Using $GM_E = V^2 a$ (from the velocity equation) to eliminate GM_E ,

$$\frac{dP}{dt} = -\frac{6\pi^2 a^3 \rho V C_D A}{m P V^2 a}$$

$$\frac{dP}{dt} = -\frac{6\pi^2 a^2 \rho C_D A}{m P V}$$

Finally, for a circular orbit the arc length traveled per period is $2\pi a = PV$,

$$\frac{dP}{dt} = -\frac{6\pi^2 a^2}{m (2\pi a)} \rho C_D A$$

$$\frac{dP}{dt} = -\frac{3\pi a \rho C_D A}{m}$$

$$\boxed{\frac{dP}{dt} = -3\pi a \rho \frac{C_D A}{m}}$$

(ii.)

(iii.)

(iv.)

Problem 6: Atmosphere Uncertainty and Neutral Winds

Problem 7: Air Breathing Thruster

Problem 8: Missions